

## uq\_bench.metrics

Targets  $y_i$  with prediction interval  $[\ell_i, u_i]$  over  $N$  test points at miscoverage level  $\alpha$ , with lower and upper quantile levels  $\alpha/2$  and  $1 - \alpha/2$ .

### picp

Prediction interval coverage probability, the fraction of targets inside the interval.

$$\text{PICP} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}\{\ell_i \leq y_i \leq u_i\}$$

### mpiw

Mean prediction interval width.

$$\text{MPIW} = \frac{1}{N} \sum_{i=1}^N (u_i - \ell_i)$$

### niw

Mean interval width normalized by the target range.

$$\text{NIW} = \frac{\text{MPIW}}{\max_i y_i - \min_i y_i}$$

### pinball

Average pinball loss of the two endpoints, with the pinball loss at level  $\tau$  defined as

$$\text{QL}_\tau(y, q) = (y - q)(\tau - \mathbb{1}\{y \leq q\}),$$

so that

$$\text{PINBALL} = \frac{1}{2N} \sum_{i=1}^N [\text{QL}_{\alpha/2}(y_i, \ell_i) + \text{QL}_{1-\alpha/2}(y_i, u_i)].$$

### aisl

Average interval score loss, the width plus a  $2/\alpha$  penalty for each target outside the interval.

$$\text{AISL} = \frac{1}{N} \sum_{i=1}^N \left[ (u_i - \ell_i) + \frac{2}{\alpha} (\ell_i - y_i) \mathbb{1}\{y_i < \ell_i\} + \frac{2}{\alpha} (y_i - u_i) \mathbb{1}\{y_i > u_i\} \right]$$

### nciw

Normalized interval width after rescaling each half-width around the center  $f_i$  by the smallest factor that reaches coverage  $1 - \alpha$ . The per-point covering scale is

$$c_i = \begin{cases} \frac{f_i - y_i}{f_i - \ell_i} & \text{if } y_i < f_i \\ \frac{y_i - f_i}{u_i - f_i} & \text{if } y_i > f_i \\ 0 & \text{otherwise} \end{cases}$$

and with  $c^*$  the  $\lceil (1 - \alpha)N \rceil$ -th smallest  $c_i$ ,

$$\text{NCIW} = c^* \cdot \text{NIW}.$$